BIKE SHARING STATION ANALYSIS – POWER BI PROJECT

1. Project Overview and Objective

This project analyses bike-sharing station information using Power BI. The report presents station availability, bike capacity, available bikes, available bike stands, and station status across cities and contracts.

The main objective is to understand the distribution and operating status of bike-sharing stations, monitor available capacity, and provide an interactive dashboard to support operational decisions.

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2. Data Sources

- Source Description: Bike-sharing station dataset imported into Power BI. The PBIX report identifies the primary table as Feuil1. The original dataset provider and collection timeline are not specified in the supplied report.
- Domain: Transportation / Urban Mobility / Bike-sharing Operations Analytics.

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3. Problem Statement

- To analyse the total number of bike-sharing stations and their distribution across cities.
- To compare open and closed station counts and understand station operating status.
- To monitor total bike capacity, available bikes, and available bike stands.
- To compare station and bike availability across cities and contract names.

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4. Attribute (Column / Features) Details

Attribute Name	Data Type	Description
Employee_ID	Integer / String	Unique identifier for each employee
Name	String (Text)	Name of the employee
Age	Numeric (Integer)	Age of the employee in years
Department	Categorical	Department where employee works
city	Categorical / Text	City in which the bike-sharing station is located.
Station	Categorical / Text	Station identifier or station name used in the report.
status	Categorical / Text	Operating status used to distinguish open and closed stations.
Contract Name	Categorical / Text	Contract or operating scheme associated with the station.
Available Bikes	Numeric	Number of bikes currently available.
Bike Stands	Numeric	Total bike stands at the station.
Available Bike Stands	Numeric	Number of bike stands currently available.
bonus	Categorical / Numeric (verify)	Bonus-related field referenced in the report; confirm its exact meaning and data type in the source.

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5. Tools & Technologies

- Power BI: Data modelling, DAX measures, report visuals, and interactive dashboard creation.

- Excel / Power Query: Can be used for source-data inspection, cleaning, transformation, and validation before loading into Power BI. The supplied PBIX alone does not confirm which cleaning steps were performed.

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6. Data Pre-Processing (Excel / Power Query)

Tasks to document / validate against the source dataset:

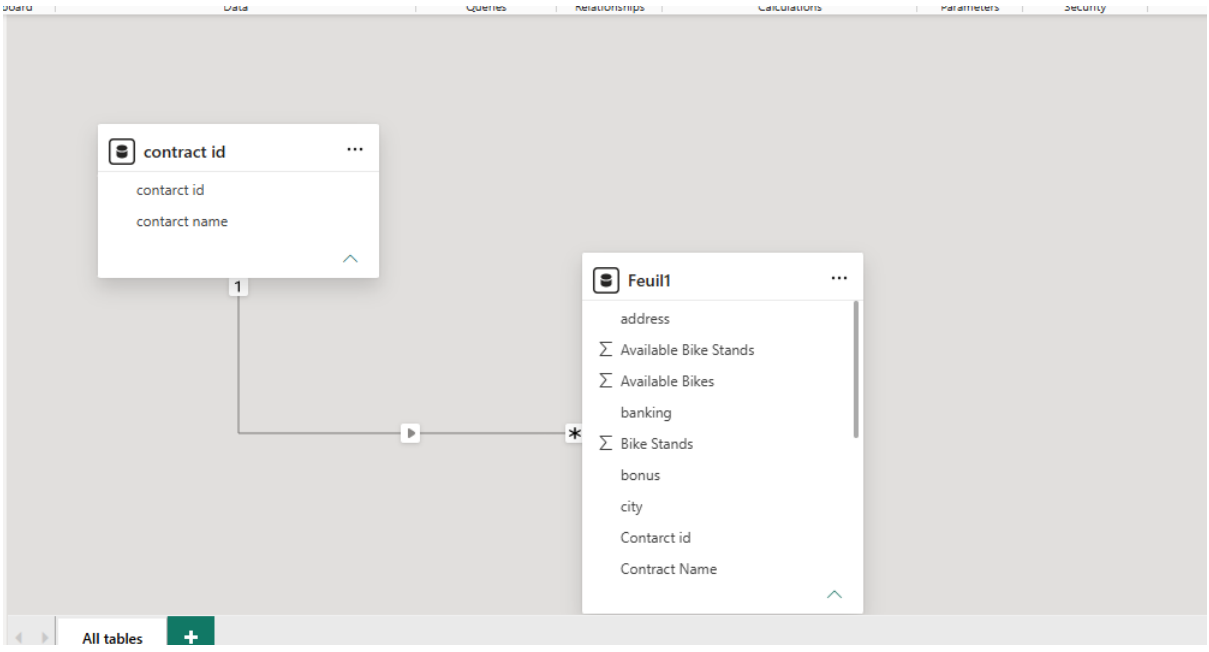
- Inspect data types for city, station, status, contract name, bonus, bike counts, and bike-stand counts. Check missing values and duplicate station records before analysis.
- Standardize city, station, and status text values. Validate numeric fields such as Available Bikes, Bike Stands, and Available Bike Stands.
- Filter and sort records for city, station status, and contract-level analysis. Validate totals against the original dataset.
- The PBIX report uses a primary table named Feuil1. A separate fact-and-dimension model is not confirmed from the supplied report definition.

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7. Data Modelling and DAX (Power BI)

- The report visuals reference the Feuil1 table. Review the Model view in Power BI Desktop to confirm relationships and cardinality; the supplied report definition does not establish a separate multi-table model.

INSERT SCREENSHOT: POWER BI MODEL VIEW



- Measures / fields referenced by the report include Total bike capcity, Total station, available total bike, Available total bike stand, closed station, and open station. Confirm each measure's DAX definition in the PBIX before including formulas in a final submission.

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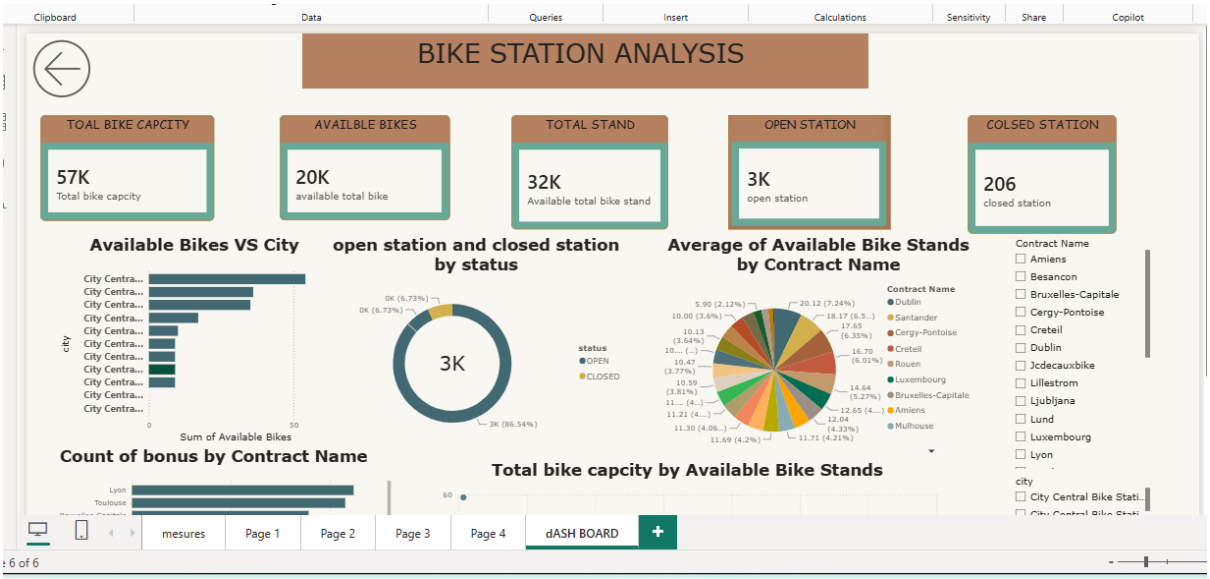
8. Analysis and Visualizations (Power BI)

Dashboard Features

- The report includes KPI cards and visual analyses using city, station, status, contract name, Available Bikes, Available Bike Stands, Bike Stands, and bonus fields. Visuals include city-wise comparisons and station-status summaries.

- Use city and status filters/slicers where available to interactively explore station availability and capacity.
- Use clear visual titles, data labels, and consistent formatting. Confirm bookmarks and drill-down Pbehaviour in Power BI Desktop before claiming they are implemented.
- The PBIX contains multiple report pages, including a page named “dASH BOARD”. Add a screenshot of the final consolidated dashboard below.

INSERT SCREENSHOT: FINAL POWER BI DASHBOARD / REPORT



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9. Insights & Conclusions

- The report is designed to show station distribution, operating status, and bike / stand availability by city and contract. Insert the actual values and observations from the dashboard after validating the filters and measures.
- Analysis insights:
  - Descriptive: Summarise total stations, total bike capacity, available bikes, available bike stands, and open / closed station counts.
  - Diagnostic: Compare station status and availability across cities and contracts to identify differences. The reasons for those differences are not provided by the report fields alone.
  - Predictive: No forecasting model is evident in the supplied PBIX. Predictive conclusions should not be claimed unless a forecasting method is added.
  - Prescriptive: Use low availability or high closed-station counts as prompts for operational review; validate thresholds and actions with the bike-sharing operator.

10. Conclusion

The Power BI report provides a visual overview of bike-sharing station capacity, availability, and operating status. It can support monitoring across cities and contracts. Final numerical findings should be added after checking the dashboard measures and source data.